1 You are going to investigate the effect of the concentration of salt solutions on osmosis in potato tissue.

Read all the instructions but DO NOT DO THEM until you have drawn a table for your results in the space provided in 1(a)(ii).

You should use the safety equipment provided while you are doing the practical work.

You are going to use three different concentrations of salt solution.

Step 1 Label three beakers **S1**, **S2** and **S3**.

(a) **(i)** Decide the volumes of salt solution **S** and distilled water **W** you need to make 50 cm^3 of a 0.5 mol per dm^3 salt solution.

Complete Table 1.1 by writing in the volumes of **S** and **W** you will use to make the 0.5 mol per dm^3 salt solution.

Table 1.1

beaker	volume of 1.0 mol per dm^3 salt solution **S** / cm^3	volume of distilled water **W** / cm^3	final concentration of salt solution / mol per dm^3
S1	50	0	1.0
S2	....	....	0.5
S3	0	50	0.0

[1]

Step 2 Use the volumes of 1.0 mol per dm^3 salt solution **S** and distilled water **W** shown in Table 1.1 to make the salt solutions in beakers **S1**, **S2** and **S3**.

Step 3 Put the potato cylinders onto a white tile and cut each potato cylinder into 3 mm discs, as shown in Fig. 1.1.

You will need 30 discs of potato for this investigation and one extra potato disc for Question **1(b)**. Leave the extra potato disc on the white tile.

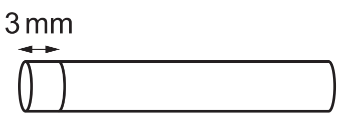

Fig. 1.1